

Kristiania University College
PGR211- Advance Programming for Data Science
Worksheet 1

1. Write a program that asks the user to enter a weight in kilogram, then converts it to pound and prints it on the screen. Note that 1 kilogram is equal to 2.2 pounds.

Sample run:

Enter a weight in kilogram: 10
10.0 kilogram(s) = 22.0 pound(s)

2. Write a program that prompts the user for a radius and then prints
 - The area and circumference of a circle with that radius
 - The volume and surface area of a sphere with that radius

Area of a circle = πr^2

Circumference of a circle = $2 \pi r$

Volume of a sphere = $\frac{4}{3} \pi r^3$

Surface area of a sphere = $4 \pi r^2$

3. Write a program that prompts the user for a username and password, then hides the password using (*) according to the length of the password and greets the user as follows.

Sample run:

Enter your username: sky_blue
Enter your password: 123@456

Hi sky_blue. Your password [*****] is 7 characters long.

4. Write a program that asks the user to input
 - The number of gallons of gas in the tank
 - The fuel efficiency in miles per gallon
 - The price of gas per gallon

Then print the cost per 100 miles and how far the car can go with the gas in the tank.

5. Write a program that reads an integer and prints whether it is negative, zero, or positive.
6. Write a program that reads in three floating-point numbers and prints the largest of the three inputs **without** using the **max** function.

7. Write a program that reads in two floating-point numbers and tests whether they are the same up to two decimal places. Here are two sample runs.

Sample run

Enter a floating-point number: 2.0
Enter a floating-point number: 1.99998
They are the same up to two decimal places.

Enter a floating-point number: 2.0
Enter a floating-point number: 1.98999
They are different.

8. Write a program to simulate a bank transaction. There are two bank accounts: checking and savings. First, ask for the initial balances of the bank accounts; consider negative initial balances as zero. Then ask for the transactions; options are deposit, withdraw, and transfer. Then ask for the account; options are checking and savings. Then ask for the amount; reject transactions that overdraw an account. At the end print the balances of both accounts.
9. Write a program using **while loop** which prints the positive odd numbers till the user input.

Sample run:

Enter a positive integer: 10
The odd number(s) till number 10 ---> 1 3 5 7 9

10. Write a program using **while loop** which prints the sequence of stars reduced by one for each line depending upon the user input.

Sample run:

Enter the number of stars you want in the first row: 6

The star sequence is

```
*****  
*****  
****  
***  
**  
*
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11. Write a program using **while loop** which prints the factorial of a positive integer given by user.
[In mathematics, the factorial of a positive integer n , denoted by $n!$, is the product of all positive integers less than or equal to n]. For example:

$$0! = 1$$

$$1! = 1$$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

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$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 3 \times 2 \times 1$$

Sample run:

Enter n: 5

5! = 120

Another sample run:

Enter n: 0

0! = 1

12. Write a program using **while loop** and **if/elif/else statements** that gets monthly sales of 5 employees from the user and then calculates and prints the income according to the following table:

Monthly Sales	Income
≥ 30000	575 + 20% of the sale
25000 – 30000 (25000 included)	520 + 10% of the sale
20000 – 25000 (20000 included)	500 + 5% of the sale
< 20000	400 + 2% of the sale

Sample run:

Enter monthly sales: 25000

Income is 3020.0

Enter monthly sales: 20000

Income is 1500.0

Enter monthly sales: 30000

Income is 6575.0

Enter monthly sales: 5000

Income is 500.0

Enter monthly sales: 26000

Income is 3120.0